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09/03/2025
CPSC 321

Homework 1:

Part A: Historical Context

1. — I found Bachman’s insight as he worked on the design of IDS in the early 1970’s to be particularly interesting and honestly pretty inspiring. The most interesting aspect is Bachman was building a navigational database system for organizing and accessing information but always envisioning the DBMS, he just needed the right understanding and people around him, but to him, it was always meant to happen. For this reason, I find the insight inspiring. Bachman was pioneering foundational concepts going far beyond simply storing or just accessing files, to think a man, within his pure elemental passion going on to learn, create, critique which slowly lead to the inception of what we know about DBMS today just makes me wonder what any of us could be really capable of.

   — Building off the previous paragraph, I think where the story got really good was when Codd and Bachman were presenting their data models in a face-to-face panel discussion at the 1974 SIGFIDET meeting. It was very interesting to picture two intellects in a somewhat legendary setting discussing their respective work but just lead to what was considered a "watershed event for the database industry" where by the time people left the conference, “the relational model had become the new mainstream for research in data management”. I realize at the time of course there was a question of viability as with anything but it's fascinating to think the future of the database industry was heavily influenced by a moment between two passionate researchers in some conference room.

   — Another thing I found really interesting in the reading is when Bachman first read Codd’s award winning paper, he was “not too impressed” and thought it was just a bunch of not very useful “mathematical jargon”. I think this is an example of a concept so ahead of its time you must see the utility before you understand why it was created, an example of this is like the early years of the internet where people really weren’t sure what to even do with it. Luckily that tutorial was shown to show its viability to Backman because maybe we would've seen a different iteration of SQL or who knows what. It is always interesting to see what types of human mistakes or oversights even brilliant minds might miss.

— The one thing I found interesting but didn't make the cut was IBM's magnetic disks technology that was used for NASA's Apollo moon landing program back in the 1960s. It's one of those things that feel like tech lore, I can't even imagine IBM intended for it to be used for a project like that and yet it contributed to the space race. I like to think of what those meeting rooms were like when designing something so foundational from the ground up at a developing time.

2. **1890:** Herman Hollerith invents punched cards for the U.S. census. This is the first big step in automating data processing, setting the stage for everything else.

   1956: IBM introduces magnetic disks (RAMAC). This was huge because it allowed direct access to data instead of slow, sequential tape reads. It opened the door for new ways to organize and query information.

   1960s: IBM builds the Information Management System (IMS) for NASA's Apollo program. I added this because it's one of the first real database management systems, proving structured data storage could support massive projects like landing on the moon.

   1969: IBM releases IMS commercially, and Charles Bachman develops IDS (Integrated Data Store). I put this in because IDS and IMS mark the start of database systems that multiple applications could share.

1970: E.F. Codd publishes his award winning paper. This is the birth of relational databases, the model that still dominates today.

1973: Charles Bachman wins the Turing Award for his IDS work. I included this because it shows early recognition of database systems as a core part of computer science.

1974: SIGFIDET conference, where Codd and Bachman debated relational vs. network data models. This was a turning point because the relational model became the new mainstream. Also, Chamberlin and Boyce quietly introduced SQL which became the database standard.

1988: System R and INGRES teams receive the ACM Software System Award. This is important because it validates their pioneering work and impact on the entire database industry.

3. The author's initial impressions of Codd's award winning 1970 paper on first reading was not too impressed, stating "the paper contained a lot of mathematical jargon" and that while it introduced the concepts of data independence and normalization, it was mainly theoretical and not grounded in practical engineering. His views changed after the Fall of 1972, when he attended the seminar at IBM Yorktown and later that year when he attended COINS-72 in Miami Beach where he met Ted Codd and Chris Date presenting a tutorial on RDBMS demonstrating its feasibility in a production environment.

   After Codd, research kept driving industry products. The article shows how System R, IBM's research project in the 1970s, directly led to SQL becoming the standard database language we still use today. What started as an experimental query language called SQL in a research lab became the foundation for practically every major database system.

Part B: Warm Up

4. I decided to focus on the Steam desktop application as I have been using it frequently since elementary school. Obviously, Steam needs to persist a whole ton of particularly valuable data/information, Hosting millions of users with constantly updating data while being done in a truly intuitive, efficient, and secure way.
   Transactions could have attributes like transactionID, paymentMethod, amount, date, productID, foreign key maybe UserID?,
   Reviews could have attributes like userID, gameID, rating, date,
   Friends could have attributes like friendID, status, level,
   Game in Library might have userID, gameID, gameCount, loggedHours, 3rd party credentials/permissions

   I'm sure there are far more complex systems with deeply embedded optimized databases used by Steam but I can imagine they use something similar to track things like games, permissions, data, store, friends, and so on.

5. OLAP: Online Analytical Processing is used to analyze large amounts of data to spot trends and patterns. The Steam store is likely so successful because of their OLAP system which helps them figure out which games to advertise to a certain demographic of people at a specific time of year. This is also seen in their Steam community market, all of which makes them an astronomical amount of money, just because they build and continue to develop a sophisticated processing system.

   OLTP: OL Transaction processing is used to handle the everyday actions that happen in real time. In this case, it can be something like sending a friend request that happens instantaneously, sending a message to a friend, perhaps starting up a game? All of which needs to be processed quickly and accurately at any given time.

   — The difference between the two is OLAP is more used for insight and prediction from the data collected while OLTP just handles several updating interactions throughout.

Part C: Relational Schemas

6.

A. Schema:

   USairports(name, city, state, elevation, faa_id)

   Primary key: faa_id

   Foreign key: N/A

   Seattle-Tacoma International | Seattle | WA | 433’ | SEA |

   Los Angeles International | Los Angeles | CA | 433’ | LAX |

   Leadville-Lake County Airport | Leadville | CO | 9934’ | KLXV |

B. Airlines(airL_id, airL_name, hub_id, foundedYear)

   Primary key: airL_id

   Foreign key: hub_id → USairports(faa_id)

   AS | Alaskan Airlines | SEA | 1932 |

   WN | Southwest Airlines | LAX | 1967 |

   UA | United Airlines | LAX | 1926 |

C. Flight(airL_id, flight_number, departure, arrival, weekly_flights)

   Primary key: (airL_id, flight_number)

   Foreign key:

   airL_id → Airlines(airL_id)

   departure → USairports(faa_id)

   arrival → USairports(faa_id)

   AS | 450 | SEA | LAX | 14 |

   WN | 1200 | LAX | GEG | 7 |

   AS | 212 | GEG | SEA | 21 |

D. FlightSegment(airL_id, flight_number, departsFrom, arrivesAt, offset)

Primary key: airL_id, flight_number, offset)

Foreign key:

(airL_id, flight_number) → Flight(airL_id, flight_number)

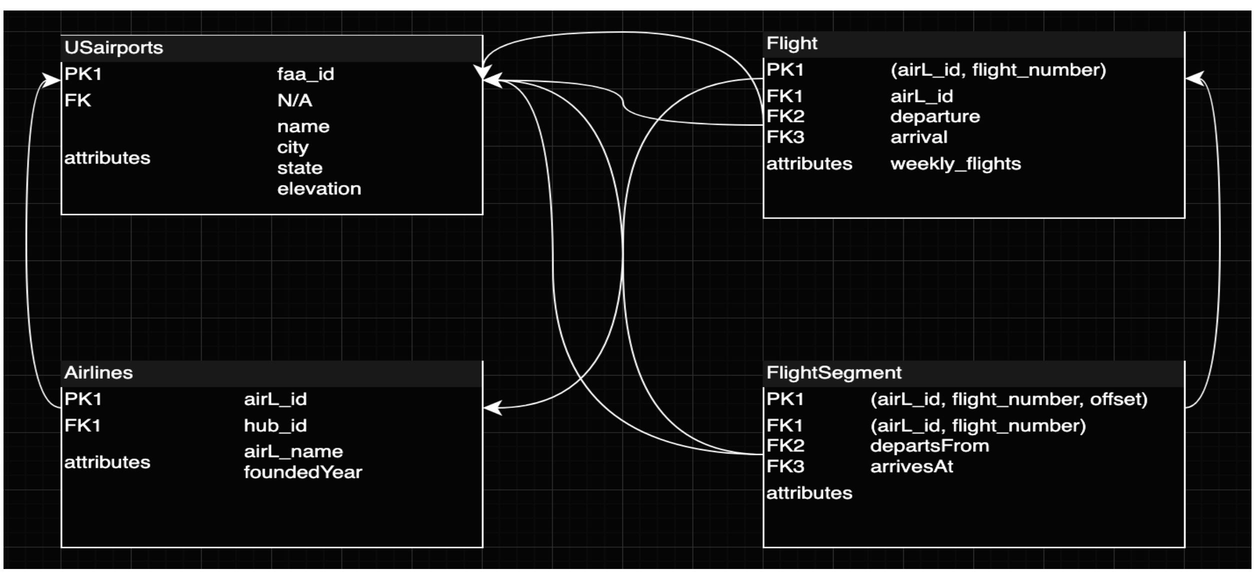
departsFrom → USairports(faa_id)

arrivesAt → USairports(faa_id)

WN | 150 | 1 | SEA | GEG |

WN | 150 | 2 | GEG | LAX |

WN | 301 | 1 | LAX | SEA |

7. 

8.

A. Schema:

Employees(employee_id, first_name, last_name, start_date, department_name)

Primary key: employee_id

Foreign key: N/A

101 | Alice | Smith | 2022-06-01 | Software Development |

102 | Bob | Jones | 2021-03-15 | Human Resources |

103 | Charlie | Brown | 2023-09-01 | Software Development |

B. Projects(project_number, sponsoring_department, Plead_id, start_date, annual_budget)

Primary key: project_number

Foreign key: Plead_id → Employees(employee_id)

P001 | Software Development | 101 | 2024-01-10 | 500000.00 |

P002 | Human Resources | 102 | 2023-05-20 | 150000.00 |

P003 | Software Development | 101 | 2025-02-01 | 1200000.00 |

C. Employees(employee_id, first_name, last_name, start_date, department_name, manager_id, manager_start_date)

Primary key: employee_id

Foreign key: manager_id -> Employees(employee_id)

101 | Alice | Smith | 2022-06-01 | Software Development |

102 | 2023-01-01 | 102 | Bob | Jones | 2021-03-15 | Human Resources | NULL | NULL |

103 | Charlie | Brown | 2023-09-01 | Software Development | 101 | 2024-06-01 |